UM-3 analysis

The x0 mass mechanism, dirac16complex as dark matter and as dark energy, and what the author may have left out

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Analysis of UM-3, 2026-10-24

Repository once-ere/Dirac_claude, folder universe_model_3

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1 How to read this document

1.1 What this document contains

The author asks for three analyses of the universe model UM-3: the mass mechanism along the hidden coordinate x_0 (the period $\pi/(6H)$ of the metric and the uncertainty relation), whether dirac16complex can be the dark matter, and whether it can be the dark energy. He also asks to be told what he may have left out. This document gives the four answers. It is self-contained: every equation it uses is written in it (with the tag it carries in every UM-3 document, FE-n for a field equation, or A-n for an equation of this document), and every symbol of every display equation is defined right after it. The sections are:

- **The answers in brief.**
- **The x_0 mass mechanism**: the period of x_0, the proper length of the hidden direction, the spectra that replace the tower of a compact dimension, and the uncertainty relation.
- **dirac16complex as dark matter**: stability, gravitation on the fold, the equation of state of the Kohn-Sham states, the relic density, direct detection, and the window of parameters that survives.

- **dirac16complex and the dark energy**: why the gas of bound states cannot be dark energy, what the dark energy of UM-3 is, and how it compares with DESI DR2.
- **What the author may have left out**: the additions that UM-3 needed, and the open problems.

1.2 The status labels

Every result carries one of the labels **AUTHOR** (from the author's notebooks `4+4.nb` and `Dirac_Einstein-Lovelock_4+4_Nash.nb` and his order of 2026-10-24), **RECORD** (from the record of this repository: the Lagrangian of dirac16complex with its canonical spin connection and contact self-interaction, the Kohn-Sham treatment), **STANDARD** (textbook physics or a published measurement, named), **ASSUMED** (a choice of UM-3 without derivation), **INVENTED** (a new interaction of UM-3: the portals of dirac16complex), **DERIVED** (computed and verified in UM-3, with the verification stated), **ESTIMATE** (an order-of-magnitude evaluation) and **OPEN** (not settled).

1.3 Every equation is followed by its symbols

Every display equation is followed by the list of EVERY symbol that appears in it, with its meaning, unit (natural units $\hbar = c = 1$), value and status. The author's coordinates are $X = \{x_0, x_1, \ldots, x_7\}$: x_0 the hidden space-like coordinate (the metric depends on it through $z = 6Hx_0$), x_1, x_2, x_3 ordinary space, x_4 the time, x_5, x_6, x_7 three extra time-like coordinates; the signature is $(+,+,+,+,-,-,-,-)$. H is the author's constant (an inverse length), a free parameter of UM-3. The Standard Model lives on the fold of the metric (the surface $z = \pi/2$, where the proper coordinate σ of the hidden direction is zero), dirac16complex and gravity in the bulk.

1.4 The answers in brief

1. **The x_0 mass mechanism.** The coordinate x_0 has the period $\pi/(6H)$, but one period is a hidden direction of infinite proper length, warped towards the fold: there is no compact dimension, no Kaluza-Klein tower with spacing $12H$, and no uncertainty bound $p_{x_0} \gtrsim 6H$. The masses that x_0 produces are of a different kind: a gap $3H$ above the massless zero mode of every massless bulk field, set by the warp, and for dirac16complex a continuum above its bulk mass m_Ψ with eight states bound to the fold at $\omega_\mathrm{b} = \sqrt{m_\Psi^2 - q_\mathrm{b}^2}$, lighter the more tightly they are bound. The uncertainty relation works in the opposite direction from the compact case: localisation lowers the energy (DERIVED).
2. **Dark matter.** The bound particles χ of dirac16complex are stable, gravitate on the fold like matter of the fold, form a pressureless gas in the dilute limit, and are a thermal relic with the measured density if they annihilate with $\langle \sigma_\mathrm{ann} \mathrm{v} \rangle = 2.2 \times 10^{-26}$ cm^3/s. Direct detection excludes this through the quark portals, allows it through the lepton portals above about 1.3 TeV, and requires the binding to be weak ($q_\mathrm{b} \lesssim 0.03\,\omega_\mathrm{b}$); the right-handed neutrinos must not be thermalised. dirac16complex can be the dark matter in this window (DERIVED, ESTIMATE).
3. **Dark energy.** No: the gas of bound states has a positive pressure that vanishes in the dilute limit, never a negative one. The dark energy of UM-3 is the vacuum energy of the fold, a free parameter with $w = -1$; it fits the CMB but keeps the 2.3-standard-deviation tension of the cosmological-constant model with the BAO of DESI DR2 (DERIVED).
4. **What was left out.** UM-3 had to add right-handed quarks and Yukawa couplings, three right-handed neutrinos (accepted by the author), a thin fluid and an Einstein-Hilbert term on the fold, a medium in the bulk that violates the null energy condition and must respond to the fold, the gauge field that the ad hoc Higgs vacuum needs, and it found several open problems (the Krein norm, the extra times, Lorentz violation in the bulk). The last section lists them.

2 The x_0 mass mechanism

2.1 The period of x_0 and the proper length of the hidden direction

$$
\begin{aligned}
g_{MN} &= \mathrm{diag}\Big(\cot^2 z,\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ -1,\\
&\qquad -\mathrm{e}^{-2a_4}b_4^2(\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4}b_4^2(\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4}b_4^2(\sin^2 z)^{1/6}\Big), \qquad z = 6Hx_0,\\
\sigma &= \mathrm{sgn}\Big(\frac{\pi}{2} - z\Big)\frac{\ln\sin z}{6H}, \qquad \mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0,\\
g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N &= \mathrm{d}\sigma^2 + \mathrm{e}^{-2H|\sigma|}\Big[\mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{e}^{-2a_4}b_4^2\big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big)\Big] - \mathrm{d}x_4^2
\end{aligned}
\tag{FE-27}
$$

Symbols in (FE-27).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4}b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

The author's metric and its proper chart. The metric depends on x_0 only through $z = 6Hx_0$, so its coefficients repeat with the coordinate period $\pi/(6H)$: one period is $0 < z < \pi$, with the fold at $z = \pi/2$ and the tips at $z = 0$ and $z = \pi$, where $g_{x_0x_0} = \cot^2 z$ diverges and the warp $(\sin^2 z)^{1/6}$ vanishes. Status: AUTHOR (the metric), DERIVED (the chart).

$$\int_0^{\pi/(6H)} \sqrt{\cot^2(6Hx_0)}\,\mathrm{d}x_0 = \frac{1}{6H}\int_0^{\pi} |\cot z|\,\mathrm{d}z = \infty, \qquad \sigma(z) = \operatorname{sgn}\Big(\frac{\pi}{2} - z\Big)\frac{\ln\sin z}{6H} \to \mp\infty \quad (z \to 0,\ \pi) \tag{A-1}$$

Symbols in (A-1).

- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no bound. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \operatorname{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- sgn: The sign function: $\operatorname{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.

The proper length of one period of x_0 is infinite: $|\cot z|$ diverges at the tips like $1/z$, so its integral diverges logarithmically, and the proper coordinate σ runs over the whole line. The hidden direction is not a circle of circumference $\pi/(6H)$ but an infinite line, warped by $\mathrm{e}^{-H|\sigma|}$ away from the fold; the tips are infinitely far away (DERIVED, `um3_fold.py`, J1).

$$p_{\mathrm{h}} \gtrsim \frac{\pi}{\pi/(6H)} = 6H, \qquad m_{\mathrm{KK}}^2 = m_\Psi^2 + \big(12H\,\mathsf{k}\big)^2 \quad (\mathsf{k} = 0, \pm1, \pm2, \dots)$$
$$\text{(for a compact direction of length } \pi/(6H)\text{, not UM-3)} \tag{A-2}$$

Symbols in (A-2).

- p_{h}: The momentum along the hidden direction of a particle confined to it, if that direction were compact (A-2, not UM-3). Unit: GeV. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- m_{KK}: The masses of the tower of modes of a field on a compact hidden direction (Kaluza-Klein masses; A-2, not UM-3). Unit: GeV. Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- k: An integer that numbers the modes of a compact direction. Unit: dimensionless. Status: STANDARD.

The mass mechanism one would expect if x_0 were a compact direction of proper length $\pi/(6H)$: a particle confined to it has, by the uncertainty relation, a momentum p_{h} along x_0 of at least about π divided by the length, $6H$, and a field periodic in x_0 has a tower of masses m_{KK} with the spacing $12H$ (the Kaluza-Klein masses, numbered by the integer k). Status: STANDARD (for a compact direction). Neither applies to UM-3, because the proper length is infinite (A-1).

2.2 What the hidden direction produces instead

$$u_{\mathrm{s}}'' + \big(\omega^2 - 9H^2\big)u_{\mathrm{s}} + 6H\,\delta_{\mathrm{D}}(\sigma)\,u_{\mathrm{s}} = 0: \qquad \omega = 0 \ \ \text{(zero mode)}, \qquad \omega \geq 3H \ \ \text{(continuum)} \tag{A-3}$$

Symbols in (A-3).

- $u_{\rm s}$: The rescaled profile of a massless scalar field of the bulk at zero momentum along ordinary space: the field is $\mathrm{e}^{3H|\sigma|}u_{\rm s}(\sigma)\,\mathrm{e}^{-\mathrm{i}\omega x_4}$. Unit: as the field. Status: DERIVED.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- $\delta_{\rm D}$: The Dirac delta function: $\int \mathrm{d}\sigma\,\delta_{\rm D}(\sigma)F(\sigma) = F(0)$. Unit: 1/length (for $\delta_{\rm D}(\sigma)$). Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

A massless field of the bulk (here a scalar at zero momentum along ordinary space, its profile written as $\mathrm{e}^{3H|\sigma|}u_{\rm s}(\sigma)$): the warp turns its mode equation into a Schrodinger problem with a constant potential $9H^2$ and an attractive delta function at the fold, which binds exactly one state, the massless zero mode; above it the continuum starts at the gap $3H$. This gap is the mass that the geometry of x_0 gives the excitations of bulk fields: it is set by H, as the author's mechanism suggests, but it comes from the warp, not from a period, and there is no tower (the spectrum above $3H$ is continuous). Status: DERIVED (`um3_bulk.py`, B1).

$$\chi' = \mathbb{K}\,\chi, \qquad \mathbb{K} = \gamma^{(0)}\big(m_\Psi + \mathrm{i}\omega\gamma^{(4)}\big), \qquad \mathbb{K}^2 = \big(m_\Psi^2 - \omega^2\big)\mathbf{1}, \qquad \chi \propto \mathrm{e}^{\pm q\sigma}, \qquad q = \sqrt{m_\Psi^2 - \omega^2} \tag{FE-37}$$

Symbols in (FE-37).

- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{3H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x} - \omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of a bound state. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- $\mathbb{K}$: The transfer matrix of the profile equation at zero momentum, $\chi' = \mathbb{K}\chi$. Unit: GeV. Status: DERIVED.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- $\mathbf{1}$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space). Unit: dimensionless. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- q: The decay rate of a profile along σ at frequency ω, $q = \sqrt{m_\Psi^2 - \omega^2}$ (imaginary in the continuum). Unit: GeV. Status: DERIVED.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

$$q_{\rm b} = -m_\Psi\left(\mu_{\rm L} + \mu_{\rm R}\right)\frac{\tanh\varsigma}{2\varsigma}, \qquad \varsigma = \sqrt{\mu_{\rm L}\mu_{\rm R}}, \qquad \omega_{\rm b} = \sqrt{m_\Psi^2 - q_{\rm b}^2}, \qquad |\chi_{\rm b}(\sigma)| \propto \mathrm{e}^{-q_{\rm b}|\sigma|} \tag{FE-40}$$

Symbols in (FE-40).

- $q_{\rm b}$: The inverse width of a state of dirac16complex bound to the fold: its profile falls as $\mathrm{e}^{-q_{\rm b}|\sigma|}$. Unit: GeV. Status: DERIVED.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- μ_X, $\mu_{\rm L}$, $\mu_{\rm R}$: The dimensionless strengths of the fold for the left-handed and right-handed halves of dirac16complex (FE-39): the jump of the profile across the fold is the exponential of $\gamma^{(0)}(\mu_{\rm L}P_{\rm L} + \mu_{\rm R}P_{\rm R})$; negative values attract. Unit: dimensionless. Status: DERIVED.
- ς: The auxiliary variable of the bound-state condition, $\varsigma = \sqrt{\mu_{\rm L}\mu_{\rm R}}$ ($\varsigma_- = \sqrt{-\mu_{\rm L}\mu_{\rm R}}$ when $\mu_{\rm L}\mu_{\rm R} < 0$). Unit: dimensionless. Status: DERIVED.

- ω_b: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.
- χ_b: The profile of a state of dirac16complex bound to the fold. Unit: GeV^2. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

dirac16complex on the static block: the continuum of its bulk modes starts at its bulk mass m_Ψ (FE-37), and the fold, if it is attractive, binds eight states below it (FE-40), with the rest energy ω_b seen on the fold and the profile $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$. Status: DERIVED (`um3_spectra.py`, S1 to S4, with the author's exact 16×16 matrices).

$$\ell_\mathrm{b} = \sqrt{\langle\sigma^2\rangle} = \frac{1}{\sqrt{2}\,q_\mathrm{b}}, \qquad m_\Psi - \omega_\mathrm{b} = m_\Psi - \sqrt{m_\Psi^2 - q_\mathrm{b}^2} \approx \frac{q_\mathrm{b}^2}{2m_\Psi} = \frac{1}{4m_\Psi\,\ell_\mathrm{b}^2} \tag{A-4}$$

Symbols in (A-4).

- ℓ_b: The width of a bound state of dirac16complex along the hidden direction: the root mean square of σ in its density $\mathrm{e}^{-2q_\mathrm{b}|\sigma|}$. Unit: length. Value: $1/(\sqrt{2}\,q_\mathrm{b})$. Status: DERIVED.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- q_b: The inverse width of a state of dirac16complex bound to the fold: its profile falls as $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$. Unit: GeV. Status: DERIVED.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- ω_b: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.

The uncertainty relation for the bound states: their width ℓ_b along the hidden direction (the root mean square of σ in the density $\mathrm{e}^{-2q_\mathrm{b}|\sigma|}$) and their mass deficit below the continuum (for weak binding). The narrower the state, the larger the deficit: localisation along x_0 makes the particle lighter, the reverse of the compact case, because the state is held by the attraction of the fold and its momentum along σ is imaginary (an evanescent wave, $p_\sigma^2 = -q_\mathrm{b}^2$ in $\omega^2 = m_\Psi^2 + p_\sigma^2$). The x_0 mass of dirac16complex is therefore a binding mass: $\omega_\mathrm{b} = \sqrt{m_\Psi^2 - q_\mathrm{b}^2}$, set by the bulk mass and by the strengths of the fold terms (the mean fields of the Standard Model at the fold), not by the period of x_0. Status: DERIVED.

2.3 The answer

The period $\pi/(6H)$ of the coordinate x_0 does not produce a mass by the uncertainty relation, because one period is an infinite proper distance. The hidden direction produces masses in two other ways: the warp gives every massless bulk field a gap $3H$ above its zero mode (a mass scale set by H, without a tower), and the fold binds dirac16complex in eight states of mass $\omega_\mathrm{b} = \sqrt{m_\Psi^2 - q_\mathrm{b}^2}$, which the uncertainty relation makes lighter the narrower they are. Status: DERIVED.

3 dirac16complex as dark matter

3.1 Stability

$$\begin{aligned}\mathcal{L}_\mathrm{P} = &-c_\Phi\left(\Phi^\dagger\Phi\right)\bar\Psi P_\mathrm{L}\Psi - \sum_X\Big[\frac{1}{4}\big(c_X^G\,G^A_{\mu\nu}G^{A\mu\nu} + c_X^W\,W^I_{\mu\nu}W^{I\mu\nu} + c_X^B\,B_{\mu\nu}B^{\mu\nu}\big)\,\bar\Psi P_X\Psi \\ &+ \frac{d_X}{2}\,B_{\mu\nu}\,\mathcal{D}_X^{\mu\nu} + \sum_f c_X^f\,J_f^\mu\,j_{X\mu}\Big],\end{aligned} \tag{IN-16}$$

$$\mathcal{D}_X^{\mu\nu} = \mathrm{i}\,\bar\Psi\,\Sigma^{\mu\nu}P_X\Psi, \qquad j_X^\mu = \mathrm{i}\big(\bar\Psi P_X D^\mu\Psi - (D^\mu\bar\Psi)P_X\Psi\big), \qquad J_f^\mu = \bar\psi_f\,\hat\gamma^\mu\psi_f$$

Symbols in (IN-16).

- $\mathcal{L}_\mathrm{P}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^\dagger \sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- G_μ^A, $G_{\mu\nu}^A$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- c_X^W: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- W_μ^I, $W_{\mu\nu}^I$: The gauge field of $SU(2)_L$ on the fold, $W_\mu^1, W_\mu^2, W_\mu^3$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- c_X^B: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_\mathrm{L}$ for $X = \mathrm{L}$, P_R for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- d_X: The dipole-portal coupling of the half X of dirac16complex with the hypercharge field strength. Unit: GeV^{-5}. Status: INVENTED.
- $\mathcal{D}_X^{\mu\nu}$: The dipole (antisymmetric tensor) density of the half X of dirac16complex at the fold, $\mathcal{D}_X^{\mu\nu} = \mathrm{i}\bar{\Psi}\Sigma^{\mu\nu}P_X\Psi$ (real, DERIVED). Unit: GeV^7. Status: DERIVED.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^ℓ and c_X^ν are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- J_f^μ: The fermion-number current of the Standard-Model multiplets of kind f, $J_f^\mu = \bar{\psi}_f \hat{\gamma}^\mu \psi_f$ summed over the three generations and the gauge components; with other labels the gauge currents (FE-16 to FE-18). Unit: GeV^3. Status: STANDARD.
- j_X^μ: The convective current of the half X of dirac16complex, $j_X^\mu = \mathrm{i}(\bar{\Psi}P_X D^\mu \Psi - (D^\mu \bar{\Psi})P_X\Psi)$ (real and joining the half with itself, DERIVED). Unit: GeV^8. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- Σ^{ab}, $\Sigma^{\mu\nu}$: The generators of Spin(4,4) on dirac16complex, $\Sigma^{ab} = \frac{1}{4}[T^a, T^b]$ (the scaled commutators of the author's matrices); $\Sigma^{\mu\nu} = \mathbf{e}_a{}^\mu \mathbf{e}_b{}^\nu \Sigma^{ab}$ along the fold. Unit: dimensionless. Status: AUTHOR.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ of IN-13; $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G_\mu^B X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- ψ, $\bar{\psi}$: A Standard-Model fermion field (a four-dimensional Weyl field of the fold with three generations, written as a four-component spinor of fixed chirality); $\bar{\psi} = \psi^\dagger\beta$; ψ_f the fields of the kind f. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^\mu = \hat{\mathbf{e}}_a{}^\mu \hat{\gamma}^{(a)}$, $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.

$$Q_\Psi = \int \mathrm{d}\sigma\, \mathrm{d}^3x\, \sqrt{|g|}\; \Psi^\dagger \mathfrak{B} \Psi, \qquad \mathfrak{B} = \mathrm{i}\,\sigma_{16}\,\gamma^{(4)}, \qquad \mathfrak{B}^\dagger = \mathfrak{B}, \qquad \mathfrak{B}^2 = \mathbf{1}, \qquad E_o = -\omega_o\, \mathcal{Q}_o \tag{FE-43}$$

Symbols in (FE-43).

- Q_Ψ: The conserved charge of dirac16complex (the Noether charge of a change of the phase of Ψ by a constant), a Krein form (indefinite). Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- x: The coordinates of the fold: x_i $(i = 1, 2, 3)$ of ordinary space, x_4 the time, x_r $(r = 5, 6, 7)$ the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathfrak{B}$: The Krein operator, $\mathfrak{B} = \mathrm{i}\sigma_{16}\gamma^{(4)}$: a Hermitian involution with the signature (8, 8); $\Psi^\dagger\mathfrak{B}\Psi$ is the charge density. Unit: dimensionless. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- σ_{16}: The matrix of the Dirac adjoint, $\sigma_{16} = T_0T_1T_2T_3$, real and symmetric; $\sigma_{16}T_a$ is real and antisymmetric. Unit: dimensionless. Status: AUTHOR.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- $\mathbf{1}$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space). Unit: dimensionless. Status: STANDARD.
- E: An energy: E_o the energy of the Kohn-Sham orbital o; in the neutrino sector the energy of a neutrino. Unit: GeV. Status: STANDARD.
- o: The label of a Kohn-Sham orbital (a single-particle state of dirac16complex); a sum over o runs over the orbitals. Status: STANDARD.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- $\mathcal{Q}_o$: The charge Q_Ψ of the Kohn-Sham orbital o (positive or negative). Unit: dimensionless. Status: DERIVED.

The portals of dirac16complex to the Standard Model (IN-16) and its conserved charge (FE-43). Every portal term, like the bulk Lagrangian and the self-interaction, contains dirac16complex only in the bilinears $\bar\Psi \cdots \Psi$: the action is invariant under a constant change of the phase of Ψ, and the charge Q_Ψ is conserved. A particle χ (a bound state of dirac16complex) can therefore annihilate only with an antiparticle, and the lightest states of dirac16complex, the bound states of the fold, cannot decay into the Standard Model: they are stable. Status: DERIVED (from the form of IN-16; the Krein signature of the charge, DERIVED in `um3_fields.py`, F3, and `um3_spectra.py`, S4).

3.2 Gravitation on the fold

$$\kappa \int_{\text{layer}} \mathcal{T}_\Psi{}^a{}_b\, \mathrm{d}\sigma = \frac{\kappa}{\mathcal{Z}\, V_*}\, \mathcal{T}_\chi{}^a{}_b \qquad \left(H \ll q_\mathrm{b}\right) \tag{A-5}$$

Symbols in (A-5).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_\Psi$: The energy-momentum tensor of dirac16complex, $\mathcal{T}_{MN} = -(2/\sqrt{|g|})\,\delta S_\Psi/\delta g^{MN}$. Unit: GeV^8. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.

- b: A frame index (as a). Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3$ at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $\mathcal{T}_\chi{}^a{}_b$: The four-dimensional energy-momentum tensor of the gas of the bound particles χ of dirac16complex, per proper volume of ordinary space at the fold (energy density ρ_χ). Unit: GeV^4. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- q_b: The inverse width of a state of dirac16complex bound to the fold: its profile falls as $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$. Unit: GeV. Status: DERIVED.

The particles χ live in the bulk, in a layer of width $1/q_\mathrm{b}$ around the fold. The junction conditions FE-4 integrate the bulk equations across the fold; if the layer is thin compared with the scale $1/H$ on which the metric varies, integrating across the whole layer gives the jump of the junction potential the additional source A-5, where $\mathcal{T}_\chi$ is the four-dimensional energy-momentum tensor of the gas of χ (per proper volume of ordinary space) and $\mathcal{Z}V_*$ the proper volume of the extra times at the fold ($\mathcal{Z} = 1$ with frozen extra times). This is exactly the form in which the matter of the fold enters ($\mathcal{S} = \mathcal{T}_\mathrm{SM}/V_* + \ldots$, FE-5): the particles χ gravitate on the fold like matter of the fold, with the same Newton constant, and enter the Friedmann equation of FE-59 as ρ_χ. The corrections are of relative order H/q_b. Status: DERIVED (from FE-4, FE-5), ESTIMATE (the corrections).

3.3 The equation of state of the Kohn-Sham states

KS_ANALYSIS_EOS

3.4 The relic density, direct detection and the window

$$\mathfrak{g}_\Phi = \frac{\partial\omega_\mathrm{b}}{\partial\mu_\mathrm{L}}\,\frac{c_\Phi}{V_*}, \qquad \mathfrak{g}_G = \sum_X \frac{\partial\omega_\mathrm{b}}{\partial\mu_X}\,\frac{c_X^G}{4V_*}, \qquad \mathfrak{g}_f = \sum_X \frac{\partial\omega_\mathrm{b}}{\partial\mu_X}\,\frac{2\omega_\mathrm{b}\,c_X^f}{V_*}, \qquad \frac{\partial\omega_\mathrm{b}}{\partial\mu_X} = -\frac{q_\mathrm{b}}{\omega_\mathrm{b}}\,\frac{\partial q_\mathrm{b}}{\partial\mu_X},$$
$$\Delta\omega_\mathrm{b} = \mathfrak{g}_\Phi\,\Delta\langle\Phi^\dagger\Phi\rangle + \mathfrak{g}_G\,\Delta\langle G^2\rangle + \sum_f \mathfrak{g}_f\,\Delta n_f, \qquad \mathfrak{g}_{h\chi} = \mathfrak{g}_\Phi\,v = -\frac{m_\Psi^2\,\mu_\mathrm{L}^2}{2\,\omega_\mathrm{b}\,v} \quad (\mu_\mathrm{R} = 0) \tag{FE-47}$$

Symbols in (FE-47).

- $\mathfrak{g}_\Phi$, $\mathfrak{g}_G$, $\mathfrak{g}_f$, $\mathfrak{g}_{h\chi}$: The four-dimensional couplings of the bound states χ of dirac16complex to the Standard Model (FE-47): to $\Phi^\dagger\Phi$, to the gluon kinetic density, to the number densities of the fermions of kind f, and the Yukawa coupling of the Higgs boson to χ. Unit: $\mathfrak{g}_\Phi$: GeV^{-1}; $\mathfrak{g}_G$: GeV^{-3}; $\mathfrak{g}_f$: GeV^{-2}; $\mathfrak{g}_{h\chi}$: dimensionless. Status: DERIVED.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- ω_b: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.
- μ_X, μ_L, μ_R: The dimensionless strengths of the fold for the left-handed and right-handed halves of dirac16complex (FE-39): the jump of the profile across the fold is the exponential of $\gamma^{(0)}(\mu_\mathrm{L}P_\mathrm{L} + \mu_\mathrm{R}P_\mathrm{R})$; negative values attract. Unit: dimensionless. Status: DERIVED.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3$ at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.

- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^ℓ and c_X^ν are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- q_b: The inverse width of a state of dirac16complex bound to the fold: its profile falls as $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$. Unit: GeV. Status: DERIVED.
- Δ: A change, difference or small variation of the quantity that follows ($\Delta\rho$ a perturbation, Δm^2 a difference of squared masses, ΔN_eff an addition to N_eff). Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $\langle G^2\rangle$: The mean square of the gluon field strength, $\langle G^A_{\mu\nu} G^{A\mu\nu}\rangle$ (the gluon condensate in the QCD vacuum, a thermal value in the plasma). Unit: GeV^4. Status: STANDARD.
- G^A_μ, $G^A_{\mu\nu}$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- n_f: The number density of the Standard-Model fermions of kind f (particles minus antiparticles, summed over generations and gauge components), the time component of the mean current $\langle J^\mu_f\rangle$ in the rest frame of the medium. Unit: GeV^3. Status: STANDARD.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV. Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.

$$\frac{\mathrm{d}n_\chi}{\mathrm{d}x_4} + 3H_4\, n_\chi = -\langle\sigma_\mathrm{ann}\mathsf{v}\rangle\big(n_\chi^2 - n_{\chi,\mathrm{eq}}^2\big), \qquad n_{\chi,\mathrm{eq}} = g_\chi\Big(\frac{\omega_\mathrm{b}\mathsf{T}}{2\pi}\Big)^{3/2}\mathrm{e}^{-\omega_\mathrm{b}/\mathsf{T}}, \qquad \Omega_\chi h_{100}^2 \simeq \frac{1.07\times10^9\ \mathrm{GeV}^{-1}\, x_\mathrm{f}}{\sqrt{g_*}\, M_\mathrm{Pl}\, \langle\sigma_\mathrm{ann}\mathsf{v}\rangle} \tag{FE-61}$$

Symbols in (FE-61).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- n_χ: The number density of the dark-matter particles χ (particles plus antiparticles). Unit: GeV^3. Status: DERIVED.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\, h_{100}$ km/s/Mpc. Status: STANDARD.
- σ_ann: The cross-section of the annihilation of χ with its antiparticle into Standard-Model particles (through the portals). Unit: GeV^{-2}. Status: DERIVED.
- v: The relative velocity of two annihilating particles; $\langle\sigma_\mathrm{ann}\mathsf{v}\rangle$ is the thermal average of the product. Unit: dimensionless. Status: STANDARD.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $n_{\chi,\mathrm{eq}}$: The number density of χ in thermal equilibrium with the plasma (non-relativistic). Unit: GeV^3. Status: STANDARD.
- g_χ: The number of internal states of the bound level of χ that are in thermal equilibrium. Unit: dimensionless. Status: DERIVED.
- ω_b: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.

- T: The temperature of the Standard-Model plasma (of the photons after the annihilation of electrons and positrons). Unit: GeV. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- Ω_χ: The density parameter of the dark matter today. Unit: dimensionless. Value: $\Omega_\chi h_{100}^2 = 0.120$ (Planck 2018). Status: DERIVED.
- h_{100}: The present expansion rate of the fold in units of 100 km/s/Mpc. Unit: dimensionless. Value: 0.674 (Planck 2018). Status: STANDARD.
- x_f: The ratio of the mass of χ to the temperature at which its annihilations stop (freeze-out). Unit: dimensionless. Value: 20 to 25. Status: STANDARD.
- g_*: The effective number of relativistic degrees of freedom at the freeze-out (g_ρ there). Unit: dimensionless. Status: STANDARD.
- M_Pl: The Planck mass, $M_\mathrm{Pl} = G_\mathrm{N}^{-1/2}$. Unit: GeV. Value: 1.221×10^{19} GeV. Status: STANDARD.

The couplings of χ to the Standard Model (FE-47) and its relic density (FE-61). Status: DERIVED, STANDARD.

requirement	consequence for UM-3	status
the measured density $\Omega_\chi h_{100}^2 = 0.120$ by thermal freeze-out	$\langle\sigma_\mathrm{ann}\mathsf{v}\rangle = 2.2 \times 10^{-26}$ cm^3/s for $\omega_\mathrm{b} = 10$ GeV to 10 TeV (FE-61 with $g_\chi = 8$, numerical solution)	DERIVED
direct detection, quark portals	a relic through quark portals scatters on nuclei 2×10^1 (10 TeV) to 5×10^9 (10 GeV) times above the LZ limit: the quark portals are too weak to set the density	ESTIMATE
direct detection, lepton portals	the scattering through a lepton loop and the photon is below the LZ limit above about 1.3 TeV	ESTIMATE
direct detection, Higgs portal	the Yukawa coupling $\mathfrak{g}_{h\chi} = -2q_\mathrm{b}^2/(\omega_\mathrm{b} v)$ of FE-47 is fixed by the binding: $q_\mathrm{b}/\omega_\mathrm{b} \lesssim 0.06$ (100 GeV) to 0.02 (10 TeV)	ESTIMATE
a thin layer and a near-luminal χ	$q_\mathrm{b} \gg H$ (A-5) and $c_\mathrm{b}^2 = q_\mathrm{b}/(q_\mathrm{b} - H)$ close to 1	DERIVED
the effective number of neutrinos (Planck 2018, ACT DR6)	the right-handed neutrinos must not be thermalised by the portal: c_X^ν much smaller than the lepton-doublet couplings	DERIVED

$$H \ll q_\mathrm{b} \lesssim 0.03\,\omega_\mathrm{b}, \qquad \omega_\mathrm{b} \gtrsim 1.3 \text{ TeV}, \qquad \mathfrak{g}_\mathrm{l} \approx \frac{1}{\omega_\mathrm{b}}\sqrt{\frac{2\pi \cdot 4\langle\sigma_\mathrm{ann}\mathsf{v}\rangle}{12}}, \qquad c_X^\nu \ll c_X^\ell \tag{A-6}$$

Symbols in (A-6).

- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- q_b: The inverse width of a state of dirac16complex bound to the fold: its profile falls as $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$. Unit: GeV. Status: DERIVED.
- ω_b: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.
- $\mathfrak{g}_\mathrm{q}$, $\mathfrak{g}_\mathrm{l}$, $\mathfrak{g}_\mathrm{p}$: The current-portal couplings $\mathfrak{g}_f$ of FE-47 taken equal for all chiral quark states ($\mathfrak{g}_\mathrm{q}$) or for all twelve chiral lepton states ($\mathfrak{g}_\mathrm{l}$), and the coupling to the proton that the lepton loop induces ($\mathfrak{g}_\mathrm{p}$). Unit: GeV^{-2}. Status: ESTIMATE.

- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- σ_{ann}: The cross-section of the annihilation of χ with its antiparticle into Standard-Model particles (through the portals). Unit: GeV^{-2}. Status: DERIVED.
- v: The relative velocity of two annihilating particles; $\langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle$ is the thermal average of the product. Unit: dimensionless. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^ℓ and c_X^ν are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.

The window in which dirac16complex is the dark matter of UM-3: a weakly bound state (q_{b} a few per cent of its mass, with the bulk scale H smaller still), heavier than about a TeV, annihilating through current portals to the twelve chiral lepton states with the coupling $\mathfrak{g}_{\mathrm{l}}$ (6×10^{-8} GeV^{-2} at 1 TeV, 6×10^{-9} GeV^{-2} at 10 TeV), with negligible quark portals and a negligible portal of the right-handed neutrinos. Status: ESTIMATE. Within it the Kohn-Sham states are those of the weakly bound point of the solution: a cold, pressureless gas. Tests: direct detection through the lepton loop at the level 10^{-47} cm^2 for a few TeV (within the reach of the next xenon experiments), annihilation into charged leptons and neutrinos today at $\langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle \approx 2\times10^{-26}$ cm^3/s (gamma rays, positrons and neutrinos from the Galactic centre and dwarf galaxies).

3.5 The answer

dirac16complex can be the dark matter of UM-3: its bound states are stable, gravitate as matter of the fold, are cold, and have the measured density as a thermal relic, provided they are weakly bound, heavier than about a TeV and leptophilic, and the right-handed neutrinos stay out of equilibrium. Status: DERIVED (stability, gravitation, equation of state, relic density), ESTIMATE (the window).

4 dirac16complex and the dark energy

$$\frac{1}{\kappa_4}\hat{G}^{\mu}{}_{\nu} = \mathcal{T}_{\mathrm{SM}}{}^{\mu}{}_{\nu} + \mathcal{T}_{\mathrm{P}}{}^{\mu}{}_{\nu} + \mathcal{T}_{\mathrm{ff}}{}^{\mu}{}_{\nu} - \frac{V_*}{\kappa}\mathcal{J}^{\mu}{}_{\nu}, \qquad \mathcal{J}^{a}{}_{b} = \kappa\,\mathcal{S}^{a}{}_{b} \text{ of FE-30},$$

$$H_4^2 = \frac{8\pi G_{\mathrm{N}}}{3}\big(\rho_{\mathrm{SM}} + \rho_\chi + \rho_\Lambda\big), \qquad 8\pi G_{\mathrm{N}} = \frac{\kappa_4}{1+5\epsilon_{\mathrm{r}}/6}, \qquad \epsilon_{\mathrm{r}} = \frac{\kappa_4 V_*}{\kappa H}, \qquad \rho_\Lambda = \rho_{\mathrm{ff}} - \frac{31HV_*}{4\kappa}, \tag{FE-59}$$

$$\frac{\hat{R}}{2\kappa_4} + q_{\mathrm{SM}} + q_\chi + q_{\mathrm{P}} + q_{\mathrm{ff}} = \frac{V_*}{\kappa}\mathcal{J}^{(r)}{}_{(r)}, \qquad q_{\mathrm{SM}} = \big\langle \mathcal{L}_{\mathrm{H}} + \mathcal{L}_{\mathrm{gauge}} + \mathcal{L}_{\mathrm{f}} + \mathcal{L}_{\mathrm{Y}}\big\rangle, \qquad q_{\mathrm{P}} = \langle\mathcal{L}_{\mathrm{P}}\rangle$$

Symbols in (FE-59).

- κ_4: The coupling of the induced gravity of the fold, $\kappa_4 = 8\pi G_{\mathrm{ind}}$; four-dimensional gravity on the fold has $8\pi G_{\mathrm{N}} = \kappa_4/(1+5\epsilon_{\mathrm{r}}/6)$ (FE-59). Unit: GeV^{-2}. Value: $\kappa_4 \simeq 8\pi G_{\mathrm{N}} = 8\pi/M_{\mathrm{Pl}}^2$. Status: ASSUMED.
- $\hat{G}^{\mu}{}_{\nu}$: The Einstein tensor of the four-dimensional metric $\hat{h}_{\mu\nu}$ of the fold. Unit: $1/\mathrm{length}^2$. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- $\mathcal{T}_{\mathrm{SM}}$: The energy-momentum tensor of the Standard Model on the fold, $\mathcal{T}_{\mathrm{H}} + \mathcal{T}_{\mathrm{gauge}} + \mathcal{T}_{\mathrm{f}}$. Unit: GeV^4. Status: STANDARD.
- $\mathcal{T}_{\mathrm{P}}$: The energy-momentum tensor of the portals of dirac16complex. Unit: GeV^4. Status: DERIVED.
- $\mathcal{T}_{\mathrm{ff}}{}^{a}{}_{b}$: The energy-momentum tensor of the fold fluid (frame components of the fold, four-dimensional normalisation). Unit: GeV^4. Status: ASSUMED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3$ at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.

- $\mathcal{J}^a{}_b$: The jump of the junction potential across the fold computed from the metric (the right-hand side of FE-30, or of FE-33 with frozen extra times): the geometric side of the junction condition, $\mathcal{J} = \kappa\mathcal{S}$. Unit: GeV. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- $\mathcal{S}^a{}_b$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); $\kappa\mathcal{S}$ is the jump of the potential $\mathfrak{F}$. Unit: GeV^7. Status: DERIVED.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_N: Newton's constant of four-dimensional gravity on the fold, $G_\mathrm{N} = 1/M_\mathrm{Pl}^2$. Unit: GeV^{-2}. Value: $6.708{\times}10^{-39}$ GeV^{-2}. Status: STANDARD.
- ρ_SM: The energy density of the Standard-Model particles. Unit: GeV^4. Status: STANDARD.
- ρ_χ: The energy density of the dark matter, the bound states χ of dirac16complex; $\rho_\chi^{(0)}$ its value today. Unit: GeV^4. Status: DERIVED.
- ρ_Λ: The vacuum energy density of the fold, the dark energy of UM-3: $\rho_\Lambda = \rho_\mathrm{ff} - 31HV_*/(4\kappa)$ (FE-59). Unit: GeV^4. Status: DERIVED.
- ϵ_r: The bookkeeping parameter of the response of the bulk to the matter of the fold: the ratio of the bulk to the fold gravity, $\epsilon_\mathrm{r} = \kappa_4 V_*/(\kappa H)$. Unit: dimensionless. Value: $\lesssim 10^{-15}$. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- ρ_ff: The energy density of the fold fluid. Unit: GeV^4. Value: $31HV_*/(4\kappa)$ at the static block. Status: ASSUMED.
- $\hat{R}$: The scalar curvature of the metric $\hat{h}_{\mu\nu}$ of the fold. Unit: 1/length^2. Status: STANDARD.
- q_SM: The pressure of the Standard Model along the extra times at the fold: the mean on-shell Lagrangian density of the Standard-Model fields (FE-5: the extra-time component of the energy-momentum tensor of every term of the fold is its Lagrangian). Unit: GeV^4. Status: DERIVED.
- q_χ: The pressure of the bound states of dirac16complex along the extra times, integrated across the fold (their on-shell Lagrangian, the self-interaction energy per volume; zero for a dilute gas). Unit: GeV^4. Status: DERIVED.
- q_P: The pressure of the portals along the extra times: their mean on-shell Lagrangian density. Unit: GeV^4. Status: DERIVED.
- q_ff: The pressure of the fold fluid along the extra times. Unit: GeV^4. Value: $-61HV_*/(8\kappa)$ at the static block. Status: ASSUMED.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $\mathcal{L}_\mathrm{H}$: The Lagrangian of the Higgs doublet on the fold. Unit: GeV^4. Status: AUTHOR.
- $\mathcal{L}_\mathrm{gauge}$: The Lagrangian of the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_\mathrm{f}$: The kinetic Lagrangian of the quarks and leptons on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_\mathrm{Y}$: The Yukawa Lagrangian of the quarks and leptons with the Higgs doublet. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_\mathrm{P}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.

The Einstein equations of the fold (FE-59): the dark energy of UM-3 is the vacuum energy of the fold $\rho_\Lambda = \rho_\mathrm{ff} - 31HV_*/(4\kappa)$, the energy of the fold fluid minus the part that the kink of the static block needs. Status: DERIVED (with the ASSUMED fold fluid).

$$\rho_\Lambda = \Omega_\Lambda\,\rho_{\rm c}^{(0)} = 0.685\cdot\frac{3\big(H_4^{(0)}\big)^2}{8\pi G_{\rm N}} = 2.5\times10^{-47}\ {\rm GeV}^4 = \big(2.2\ {\rm meV}\big)^4 \tag{A-7}$$

Symbols in (A-7).

- ρ_Λ: The vacuum energy density of the fold, the dark energy of UM-3: $\rho_\Lambda = \rho_{\rm ff} - 31HV_*/(4\kappa)$ (FE-59). Unit: GeV^4. Status: DERIVED.
- Ω_Λ: The density parameter of the dark energy (the vacuum energy of the fold) today. Unit: dimensionless. Value: 0.685 (Planck 2018). Status: DERIVED.
- $\rho_{\rm c}$: The critical density, the energy density of a spatially flat fold at the expansion rate H_4; $\rho_{\rm c}^{(0)}$ its value today. Unit: GeV^4. Value: $\rho_{\rm c}^{(0)} = 1.0537\times10^{-5}\,h_{100}^2$ GeV/cm^3. Status: STANDARD.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $G_{\rm N}$: Newton's constant of four-dimensional gravity on the fold, $G_{\rm N} = 1/M_{\rm Pl}^2$. Unit: GeV^{-2}. Value: 6.708×10^{-39} GeV^{-2}. Status: STANDARD.

The vacuum energy that the measurements require ($H_4^{(0)} = 67.36$ km/s/Mpc $= 1.44\times10^{-42}$ GeV). Status: STANDARD.

Can dirac16complex be the dark energy? A dark energy needs a pressure close to minus its energy density. The gas of bound states has the pressure $P = nE_{\rm int}/N \geq 0$ of its repulsive self-interaction, which vanishes in the dilute limit (KS_DE_EOS); an attractive self-interaction gives a negative pressure but no stable state (in the saturating regime the Kohn-Sham equations have no bound solution for $\lambda_\Psi < 0$). The gas of χ is dark matter, not dark energy. The other contribution of dirac16complex to the vacuum energy, the energy of its unoccupied and its negative-energy states (the sea, which the Kohn-Sham states leave out), is a renormalisation of ρ_Λ that UM-3 does not compute; its sign and size are OPEN, and with the indefinite Krein norm it is not even clear that it is bounded. The dark energy of UM-3 is therefore the vacuum energy of the fold, a free parameter: its value (2.2 meV)^4 is accommodated by choosing $\rho_{\rm ff}$, not explained, and it has $w = -1$ exactly (to order $\epsilon_{\rm r} \lesssim 10^{-15}$). With the Planck 2018 parameters UM-3 reproduces the CMB, but it keeps the tension of the cosmological-constant model with the BAO of DESI DR2 (2.3 standard deviations) and cannot produce the evolving dark energy ($w_0 > -1$, $w_a < 0$) that DESI DR2 with the CMB prefers at 3.1 standard deviations. Status: DERIVED (`um3_kohnsham.py`, K5, K7; `um3_cosmology.py`, C5), OPEN (the sea).

4.1 The answer

No: dirac16complex is not the dark energy of UM-3. Its bound states are cold dark matter; the dark energy is the vacuum energy of the fold, a free parameter with $w = -1$.

5 What the author may have left out

The author asked to be told if something important was missing from his specification. UM-3 had to add the following, and the analysis found the following problems.

item	why it is needed or what it means	where it stands
right-handed quark singlets and the Yukawa couplings	the quark doublets alone have no masses and the hypercharges cancel the gauge anomalies only with the singlets	added (the author's default, 17:20)
three right-handed neutrinos	Dirac neutrino masses and the mixing that flavour conversion needs; the anomalies still cancel	added (accepted by the author, 17:32); they must never thermalise

item	why it is needed or what it means	where it stands
the place of the Standard Model	in the bulk, every species would move with its own limiting speed and three neutrino profiles would oscillate wrongly; on the fold all share one light cone	the Standard Model lives on the fold
a thin source at the fold	the kink of the warp $\mathrm{e}^{-H\lvert\sigma\rvert}$ requires one (the fold fluid); its energy above the static value is the dark energy	ASSUMED
an Einstein-Hilbert term of the fold	bulk gravitons move 2.74 times faster than light at the static block and have no time-kinetic term at the author's vacuum; gravitational waves move with the speed of light to 10^{-15}	ASSUMED (induced gravity)
a medium of the bulk	the static late-time bulk and the frozen extra times need bulk matter that violates the null energy condition	ASSUMED, no field theory given
the response of that medium	without it the radion condition changes the expansion law (dust as $x_4^{5/6}$)	ASSUMED (the radion mode)
frozen extra times in the late universe	otherwise the couplings of the Standard Model change as $\mathrm{e}^{3Hx_4/2}$	required
a gauge field for the ad hoc Higgs vacuum	the literal ad hoc vacuum carries an isospin charge that the Gauss law forbids; with the pure-gauge $W_{x_4}^2$ it is a rotated vacuum with one radial residual, and it must decay before nucleosynthesis	DERIVED
the Krein norm of dirac16complex	the signature $(4,4)$ of the author's matrices makes the charge indefinite: half of the bound states carry negative energy	OPEN (stability)
the extra times	fields that depend on them are tachyonic	ASSUMED (no dependence), OPEN
Lorentz violation in the bulk	the particles χ move faster than light on the fold unless $q_\mathrm{b} \gg H$	a condition on the parameters
direct detection	the Higgs portal fixed by the binding and the quark portals are strongly constrained	a weakly bound, leptophilic χ above a TeV
the dark energy	a vacuum energy with $w = -1$; DESI DR2 prefers an evolving one	OPEN (tension at 2.3 to 3.1 standard deviations)
the primordial era	the transition from the author's vacuum to the late universe and the primordial spectrum are not computed	OPEN (Planck inputs)
the lightest neutrino mass and the Dirac phase	not fixed by the specification	ASSUMED: $m_1 = 0$, phase 0